REFINED DVR CRITERION FOR SEPARATEDNESS WITH DENSE IMAGE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3019-3037 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 3039-3086, the complete author proof (46 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 30 of queue_stacks-stacks-more-morphisms.json (pb=1846, stmt 3019-3037, proof 3039-3086).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-refined-valuative-criterion-separated' also exists in stacks-limits, stacks-spaces-limits — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma lemma-refined-valuative-criterion-proper: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2909–2936. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CLC: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3086–3086. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E8L: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3086–3086. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 04XS: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3086–3086. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ and $h : \mathcal{U} \to \mathcal{X}$ be morphisms of algebraic stacks. Assume that $\mathcal{Y}$ is locally Noetherian, that f is locally of finite type and quasi-separated, that h is of finite type, and that the image of $|h| : |\mathcal{U}| \to |\mathcal{X}|$ is dense in $|\mathcal{X}|$. If given any 2-commutative diagram*

$$\begin{array}{ccccc} \mathrm{Spec}(K) & \xrightarrow{u} & \mathcal{U} & \longrightarrow & \mathcal{X} \\ {\scriptstyle j}\downarrow & \nearrow_{h} & & & \downarrow{\scriptstyle f} \\ \mathrm{Spec}(A) & & \xrightarrow{y} & & \mathcal{Y} \end{array}$$

where A is a discrete valuation ring with field of fractions K and $\gamma : y \circ j \to f \circ h \circ u$, the category of dotted arrows is either empty or a setoid with exactly one isomorphism class, then f is separated.

Proof. We have to prove Δ is a proper morphism. Assume first that Δ is separated. Then we may apply Lemma lemma-refined-valuative-criterion-proper to the morphisms $\mathcal{U} \to \mathcal{X}$ and $\Delta : \mathcal{X} \to \mathcal{X} \times_{\mathcal{Y}} \mathcal{X}$. Observe that Δ is quasi-compact as f is quasi-separated. Of course Δ is locally of finite type (true for any diagonal morphism, see Morphisms of Stacks, Lemma 04XS). Finally, suppose given a 2-commutative diagram

$$\begin{array}{ccccc} \mathrm{Spec}(K) & \xrightarrow{u} & \mathcal{U} & \longrightarrow & \mathcal{X} \\ {\scriptstyle j}\downarrow & \nearrow_{h} & & & \downarrow{\scriptstyle \Delta} \\ \mathrm{Spec}(A) & & \xrightarrow{y} & & \mathcal{X} \times_{\mathcal{Y}} \mathcal{X} \end{array}$$

where A is a discrete valuation ring with field of fractions K and $\gamma : y \circ j \to \Delta \circ h \circ u$. By Morphisms of Stacks, Lemma 0E8L and the assumption in the lemma we find there exists a unique dotted arrow. This proves the last assumption of Lemma lemma-refined-valuative-criterion-proper holds and the result follows.

In the general case, it suffices to prove Δ is separated since then we'll be back in the previous case. In fact, we claim that the assumptions of the lemma hold for

$$\mathcal{U} \to \mathcal{X} \quad \text{and} \quad \Delta : \mathcal{X} \to \mathcal{X} \times_{\mathcal{Y}} \mathcal{X}$$

Namely, since Δ is representable by algebraic spaces, the category of dotted arrows for a diagram as in the previous paragraph is a setoid (see for example Morphisms of Stacks, Lemma 0CLC). The argument in the preceding paragraph shows these categories are either empty or have one isomorphism class. Thus Δ is separated. □